

POLYNOMIALS

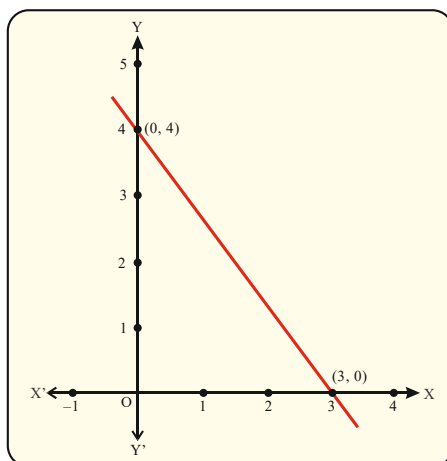
Chapter 2



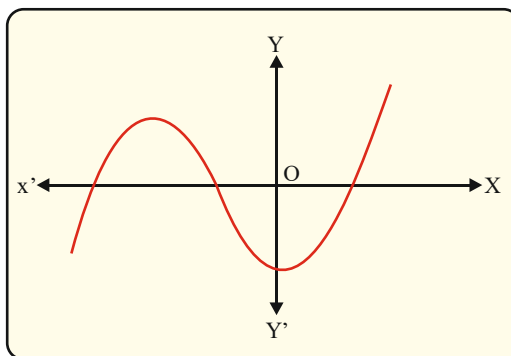
Exercise-1

Part - 1 (Basic Multiple Choice Questions)

1. The maximum number of zeroes that a polynomial of degree 3 can have is [CBSE 2012]
(a) One (b) Two (c) Three (d) None
2. The given linear polynomial $y = f(x)$ has [CBSE SQP (Standard) 2023-24]



- (a) 2 zeroes (b) 1 zero and the zero is '3'
(c) 1 zero and the zero is '4' (d) No zero
3. The graph of a polynomial is shown in figure, then the number of its zeroes is [CBSE (Basic) 2020]



- (a) 3 (b) 1 (c) 2 (d) 4
4. If -2 is a zero of the polynomial $P(x) = 3x^2 + 4x + 2k$, then the value of k is
(a) 1 (b) -1 (c) 2 (d) -2

5. The zeroes of the quadratic polynomial $x^2 + 99x + 127$ are [CBSE 2010]
 (a) Both positive (b) Both negative
 (c) One positive and one negative (d) Both equal
6. The zeroes of the quadratic polynomial $x^2 + kx + k, k \neq 0$
 (a) Cannot both be positive (b) Cannot both be negative
 (c) Are always unequal (d) Are always equal.
7. If the zeroes of the quadratic polynomial $ax^2 + bx + c, c \neq 0$ are equal, then
 (a) c and a have opposite, signs (b) c and b have opposite signs
 (c) c and a have the same sign (d) c and b have the same sign
8. If the sum of zeroes of the polynomial $x^2 - (k + 3)x + (5k - 3)$ is equal to one-fourth of the product of the zeroes, then the value of k is
 (a) 3 (b) 5 (c) 15 (d) 18
9. The quadratic polynomial, the sum of whose zeroes is -5 and their product is 6 , is [CBSE (Standard) 2020]
 (a) $x^2 + 5x + 6$ (b) $x^2 - 5x + 6$ (c) $x^2 - 5x - 6$ (d) $-x^2 + 5x + 6$
10. A quadratic polynomial having zeroes $-\sqrt{\frac{5}{2}}$ and $\sqrt{\frac{5}{2}}$ is [CBSE SQP (Standard) 2024-25]
 (a) $x^2 - 5\sqrt{2}x + 1$ (b) $8x^2 - 20$ (c) $15x^2 - 6$ (d) $x^2 - 2\sqrt{5}x - 1$
11. If a and b are zeroes of the polynomial $2t^2 - 4t + 3$, then the value of $a^2b + ab^2$ is :
 (a) $\frac{3}{4}$ (b) 2 (c) 3 (d) 4
12. While observing a basketball player in action, Pratham wondered if the path traced by the ball during its journey from the player's hands to the basket is that of a parabola. On reaching home, he traced the path on a graph paper and was happy to learn that it indeed resembles the shape of a parabola, which is the graph of a quadratic polynomial. He thought that he can also evaluate the zeroes of this polynomial.



If the zeroes of the quadratic polynomial $x^2 + (a + 1)x + b$ are 2 and -3 , then:

- (a) $a = -7, b = -1$ (b) $a = 5, b = -1$ (c) $a = 2, b = -6$ (d) $a = 0, b = -6$

13. If α, β are the zeroes of the quadratic polynomial $p(x) = x^2 - (k+6)x + 2(2k-1)$, then the value of k , if $\alpha + \beta = \frac{1}{2}\alpha\beta$ is : [CBSE Term-1 Std. 2021]
(a) -7 (b) 7 (c) -3 (d) 3
14. If α, β are zeroes of the polynomial $x^2 - 1$, then value of $(\alpha + \beta)$ is : [CBSE 2023]
(a) 2 (b) 1 (c) -1 (d) 0

Part - 2 (Basic Subjective Type Questions)

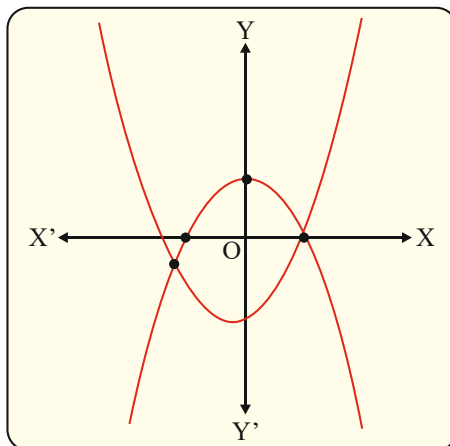
1. Find out the degrees of following polynomials.
(a) $p(x) = 7x + 5x^2 - \sqrt{3}$ (b) $q(x) = 5x^4 - 32x^2 + 5x - 8$
(c) $r(x) = x^3 - x^6 - 5\sqrt{2}$ (d) $h(x) = \frac{1}{2} - 3x$
2. Find the zeroes of the quadratic polynomial $6x^2 - 3 - 7x$ and verify the relationship between the zeroes and the coefficients.
3. If zeroes of the polynomial $x^2 + 4x + 2a$ are α and $\frac{2}{\alpha}$, then find the value of a .
4. If α, β are zeroes of quadratic polynomial $5x^2 + 5x + 1$, find the value of
(i) $\alpha^2 + \beta^2$ (ii) $\alpha^{-1} + \beta^{-1}$
5. If α and β are zeroes of a polynomial $6x^2 - 5x + 1$ then form a quadratic polynomial whose zeroes are α^2 and β^2 . [CBSE SQP 2024]
6. If m and n are zeroes of the polynomial without factorising the polynomial. [CBSE Question Bank 2023]
($3x^2 - x - 2$), find the values of the following.
(a) $\frac{1}{m} + \frac{1}{n}$ (b) $m^2 + n^2$



Exercise-2

Part - 1 (Advanced Multiple Choice Questions)

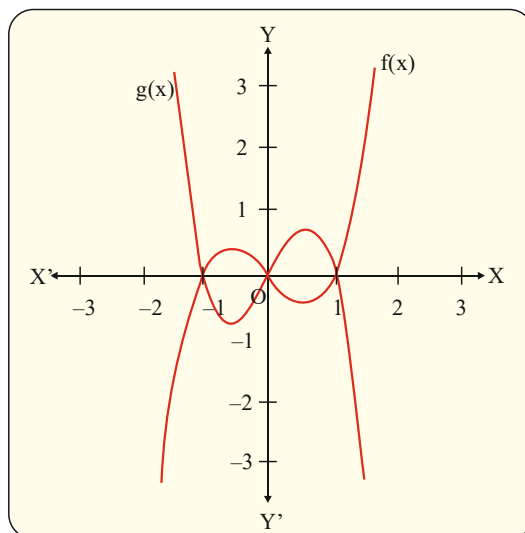
1. Two polynomials are shown in the figure below :



The number of zeroes that are common to both the polynomials is

- (a) 0 (b) 1 (c) 2 (d) 3
2. If one zero of a polynomial $3x^2 - 8x + 2k + 1$ is seven times the other, then the value of k is
- (a) $\frac{2}{3}$ (b) $-\frac{2}{3}$ (c) $\frac{3}{2}$ (d) $-\frac{3}{2}$
3. If one zero of the quadratic polynomial $f(x) = 4x^2 - 8kx + 8x - 9$ is negative of the other, then sum of the zeroes of $kx^2 + 3kx + 2$ is
- (a) 2 (b) -2 (c) 3 (d) -3
4. If α and β are the zeroes of the polynomial $f(x) = x^2 - x - k$ such that $\alpha - \beta = 9$, then the value of k is
- (a) 20 (b) 25 (c) 35 (d) 40
5. A quadratic polynomial, whose zeroes are -3 and 4, is
- (a) $x^2 - x + 12$ (b) $x^2 + x + 12$ (c) $\frac{x^2}{2} - \frac{x}{2} - 6$ (d) $2x^2 + 2x - 24$

6. Shown below are the graphs of two cubic polynomials $f(x)$ and $g(x)$. Both polynomials have the zeroes (-1) , 0 and 1 .



Anya said, “Both the graphs represent the same polynomial $f(x) = g(x) = (x+1)(x-0)(x-1)$ as they have the same zeroes”.

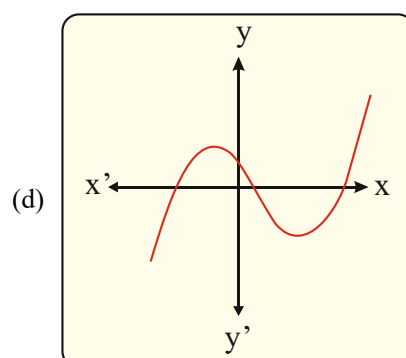
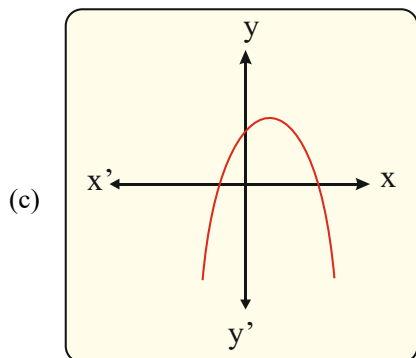
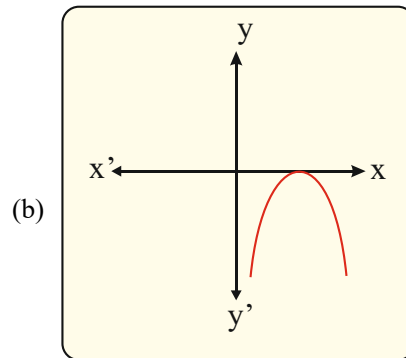
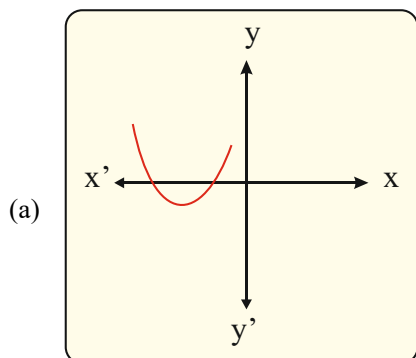
Pranit said, “Both the graphs represent two different polynomials, $f(x) = (x+1)(x-0)(x-1)$ and $g(x) = -(x+1)(x-0)(x-1)$ and only two such polynomials exist that can have the zeroes (-1) , 0 and 1 ”.

Aadar said, “Both the graphs represent two different polynomials and infinitely many such polynomials exist that have the zeroes (-1) , 0 and 1 ”.

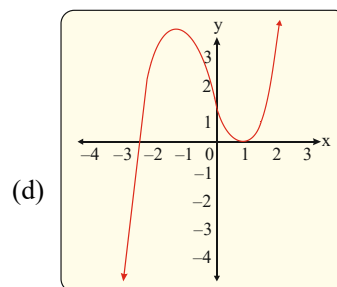
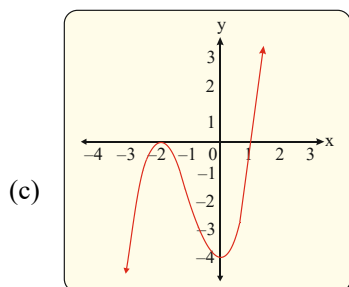
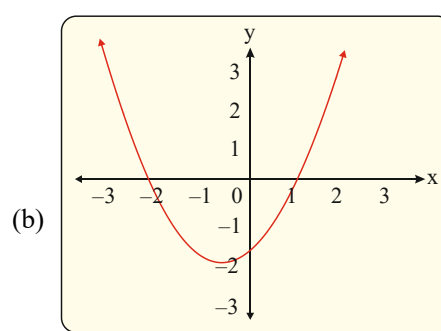
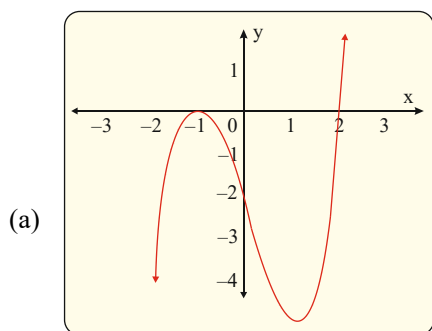
Who is right?

- (a) Anya (b) Pranit (c) Aadar (d) None of them
7. If $x^2 - 4$ is the factor of $2x^3 + k_1x^2 + k_2x + 12$, where k_1, k_2 are constant, then the value of $k_1 + k_2$ is [NTSE]
(a) 11 (b) 5 (c) -11 (d) -5
8. If $x = 3 + 3^{2/3} + 3^{1/3}$, then the value of [JSTSE]
(a) 1 (b) 0 (c) -1 (d) 2
9. Let $P(x)$ be a polynomial of degree 3 and $P(n) = \frac{1}{2}$ for $n = 1, 2, 3, 4$. Then the value of $P(5)$ is [NTSE]
(a) 0 (b) $\frac{1}{5}$ (c) $-\frac{2}{5}$ (d) $\frac{3}{5}$
10. If $f(x) = (\lambda^2 + 4)x^2 + 13x + 4\lambda$ and one zero of the polynomial is reciprocal of the other, then $\lambda =$
(a) -1 (b) 1 (c) 2 (d) -2
11. If one of the zeroes of a quadratic polynomial of the form $x^2 + ax + b$ is the negative of the other, then it:
(a) has no linear term and the constant term is negative
(b) has no linear term and the constant term is positive
(c) can have a linear term but the constant term is negative
(d) can have a linear term but the constant term is positive

12. Which of the following is not the graph of a quadratic polynomial?



13. Which of the following could be the graph of the polynomial $(x - 1)^2(x + 2)$? [CBSE Practice Set-1 2023]



16. If $\alpha - \beta$ is a zero of the polynomials $x^2 + sx + t$ and $x^2 + ux + v$ what is the relationship between α and β ?

(a) $\alpha = \frac{t-s}{v-u} + \beta$

(b) $\alpha = \beta + \frac{v-t}{s-u}$

(c) $\alpha = \frac{t-s}{v-u}$

(d) $\alpha = \frac{s-t}{v-u} - \beta$

17. Which of the following statements is true regarding the shape of the graph of $y = ax^2 + bx + c$?
- (a) The graph opens upwards if $a > 0$ (b) The graph opens downwards if $a > 0$
 (c) The graph opens upwards if $a < 0$ (d) The graph opens downwards if $a = 0$
18. What will be the value of p , if one zero of $x^2 + (p - 7)x - 25 = 0$ is the additive inverse of the other?
- (a) 4 (b) -4 (c) -3 (d) 7

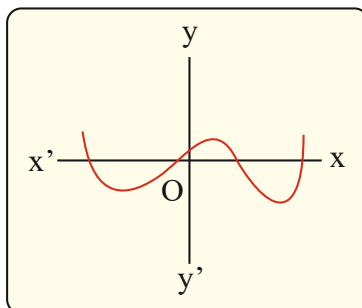
Part - 2 (Advanced Subjective Type Questions)

1. If α and β are the zeroes of the quadratic polynomial $f(x) = ax^2 + bx + c$ then evaluate $\frac{\alpha^2}{\beta^2} + \frac{\beta^2}{\alpha^2}$
2. Find the polynomial having $5 \pm \sqrt{3}$ as its zeroes.
3. A teacher asked 10 of his students to write a polynomial in one variable on a paper and then to handover the paper. The following were the answers given by the students :
- $2x + 3, 3x^2 + 7x + 2, 4x^3 + 3x^2 + 2$
 $x^3 + \sqrt{3}x + 7, 7x + \sqrt{7}, 5x^3 - 7x + 2$
 $2x^2 + 3 - \frac{5}{x}, 5x - \frac{1}{2}, ax^3 + bx^2 + cx + d, x + \frac{1}{x}$
- Answer the following questions :
- (i) How many of the above ten, are not polynomials?
 (ii) How many of the above ten, are quadratic polynomials?
4. Find the zeroes of the quadratic polynomial $f(x) = ax^2 + (b^2 - ac)x - bc$, Also verify the relationship between the zeroes and its coefficients.
5. If α and β are the zeroes of the polynomial $x^2 + 4x + 3$, find the polynomial whose zeroes are $1 + \frac{\beta}{\alpha}$ and $1 + \frac{\alpha}{\beta}$.
6. If α and β are the zeroes of the quadratic polynomial $f(x) = 2x^2 - 5x + 7$, find a polynomial whose zeroes are $2\alpha + 3\beta$ and $3\alpha + 2\beta$.
7. Find a quadratic polynomial whose zeroes are reciprocals of the zeroes of the polynomial
- $f(x) = ax^2 + bx + c, a \neq 0, c \neq 0$

[CBSE 2020]

Part - 3 (Advanced Miscellaneous Questions)

- Directions :** In the following questions, a statement of assertion (A) is followed by a statement of reason (R). Mark the correct choice as:
 - Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A).
 - Both assertion (A) and reason (R) are true but reason (R) is not the correct explanation of assertion (A).
 - Assertion (A) is true but reason (R) is false.
 - Assertion (A) is false but reason (R) is true.
- Assertion (A) :** If the graph of a polynomial intersects the X-axis at only one point, it cannot be a quadratic polynomial.
Reason (R) : A polynomial $p(x)$ of degree n can have at most n real zeroes.
- Assertion (A) :** The highest exponent of the variable in the algebraic expression is called the degree of a polynomial.
Reason (R) : The degree of $2x - 5$ is 1.
- Assertion :** The graph $y = f(x)$ is shown in the figure, for the polynomial $f(x)$. The number of zeroes of $f(x)$ is 4.
Reason (R) : The number of zeroes of the polynomial $f(x)$ is the number of points at which $f(x)$ cuts or touches the x-axis.

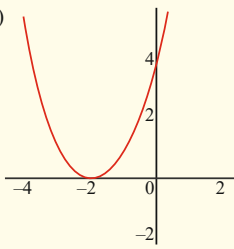
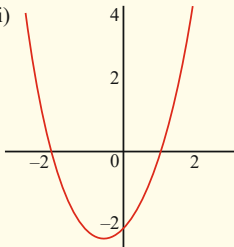
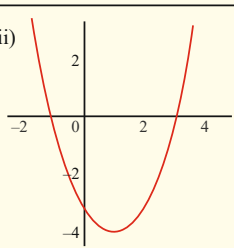
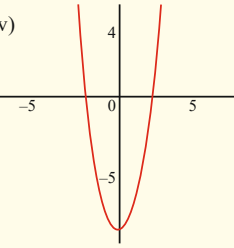


- Assertion :** Zeroes of $f(x) = x^2 - 4x - 5$ are 5, -1
Reason : The polynomial whose zeroes are $2 + \sqrt{3}$ and $2 - \sqrt{3}$ is $x^2 - 4x + 7$
- Assertion :** $(2 - \sqrt{3})$ is one of the zero of the quadratic polynomial, then the other zero will be $(2 + \sqrt{3})$.
Reason (R) Irrational zeroes (roots) always occur in pairs.
- Assertion :** The quadratic polynomial with zeros $8/9$ and 0 is $9x^2 - 8$.
Reason : If the zeros of a quadratic polynomial are p and q , then the factored form of the quadratic polynomial can be written as $p(x) = (x - p)(x - q)$.
- If p and q are the zeroes of polynomial $p(x) = x^2 - 4x + 1$, find the values of expressions given in Column I and match with values in Column II.

Column I	Column II
(a) $p+q - \frac{2}{pq}$	(i) 3
(b) $\frac{1}{p} + \frac{1}{q} - pq$	(ii) 14
(c) $3p + 3q + 2pq$	(iii) 2
(d) $5pq - 2p - 2q$	(iv) -3

- (a) A-ii, B-iii, C-iv, D-i
- (b) A-iii, B-iv, C-i, D-ii
- (c) A-iv, B-i, C-ii, D-iii
- (d) A-iii, B-i, C-ii, D-iv

8. The quadratic polynomials are mentioned in Column I. Match the following with curves in Column II.

Column - I	Column - II
(a) $2x^2 - 8$	(i) 
(b) $x^2 - 2x - 3$	(ii) 
(c) $x^2 + 4x + 4$	(iii) 
(d) $x^2 + x - 2$	(iv) 

- (a) A-iv, B-iii, C-i, D-ii
- (b) A-iii, B-i, C-iv, D-ii
- (c) A-ii, B-iii, C-iv, D-i
- (d) A-iv, B-iii, C-ii, D-i



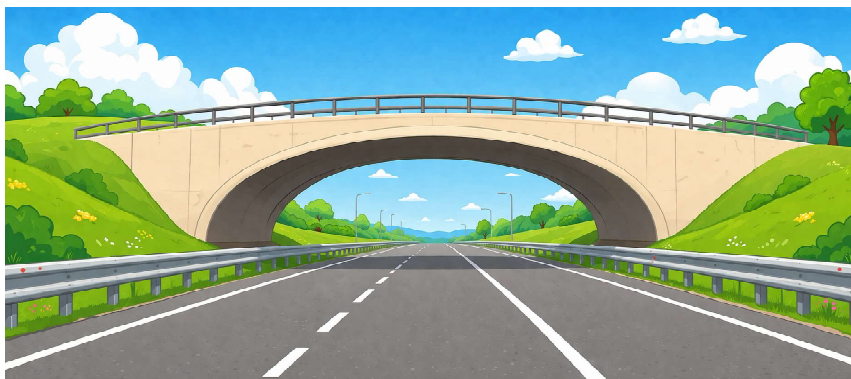
Exercise-3

(Case Based Questions)

1. Applications of Parabolas :

Highway Overpasses/Highway Underpasses

A highway underpass is parabolic in shape

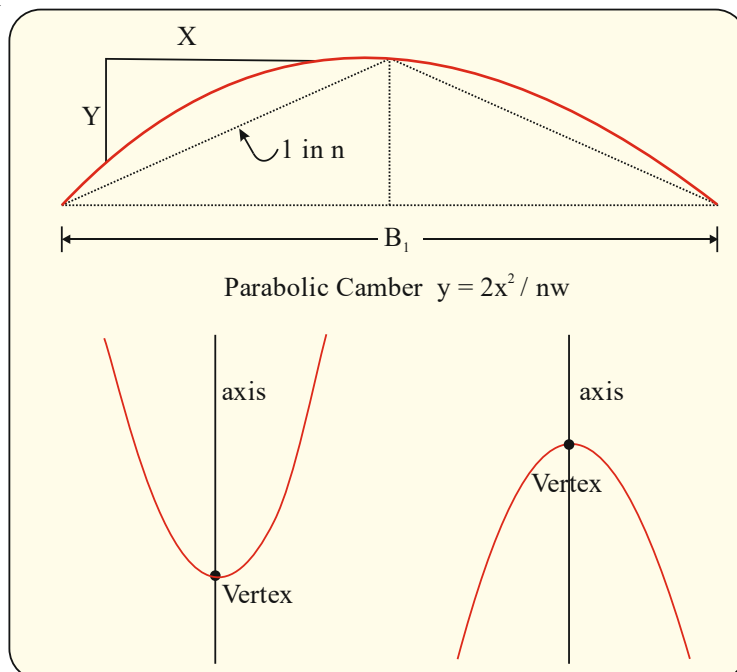
**Parabola**

A parabola is the graph that results from $p(x) = ax^2 + bx + c$.

Parabolas are symmetric about a vertical line known as the Axis of Symmetry.

The Axis of Symmetry runs through the maximum or minimum point of the parabola which is called the vertex.

Shape of Cross slope.



- (i) If the highway overpass is represented by $x^2 - 2x - 8$. Then find its zeroes.
- (ii) The highway overpass is represented graphically. Zeroes of a polynomial can be expressed graphically. Number of zeroes of polynomial is equal to number of points where the graph of polynomial.
 - (a) Intersects X-axis
 - (b) Intersects Y-axis
 - (c) Intersects Y-axis or X-axis
 - (d) None of the above

(iii) Find the representation of Highway Underpass whose one zero is 6 and sum of the zeroes is 0.

OR

(iii) The number of zeroes that polynomial $f(x) = (x - 2)^2 + 4$ can have is

2. The revenue (in Rs) of a firm is represented by the polynomial $R(x) = 5x^3 + 4x^2 + 7$, and the expenditure (in Rs) by the firm is represented by the polynomial $E(x) = 3x^3 + 2x - 1$ where x is the number of items produced by the firm in a year.

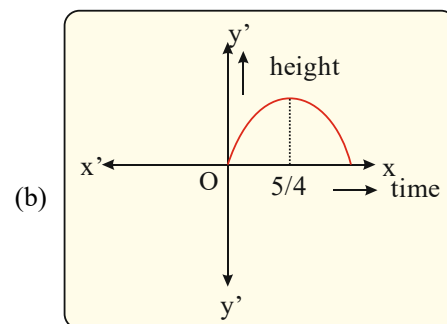
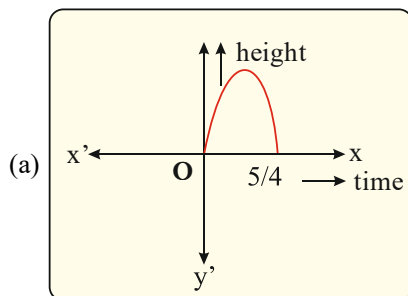
[CBSE Competency Focused Practice Questions 2024-25]

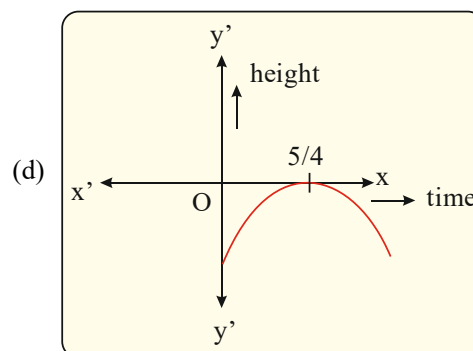
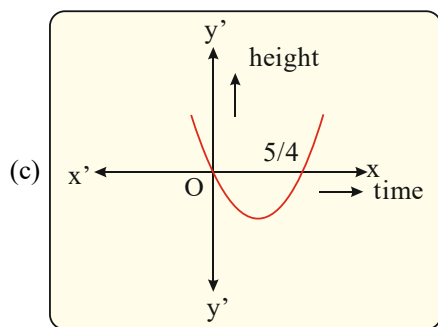
- Find the profit polynomial $P(x)$. Show your work.
 - If the firm produces 100 products in a year, find the revenue and profit (in Rs) for the firm using the polynomials. Show your work.
 - Tax is calculated on the profit using the polynomial $T(y) = 0.3y + 100$, where y represents the profit earned. Determine the tax amount (in Rs) to be paid on the profit generated from 10 items. Show your work.
3. In a pool at an aquarium, a dolphin jumps out of the water travelling at 20cm per second. Its height above water level after seconds is given by $h = 20t - 16t^2$



Based on the above, answer the following questions.

- Find zeroes of polynomial $p(t) = 20t - 16t^2$
- Which of the following types of graph represents $p(t)$?





(iii) What would be the value of h at $t = \frac{3}{2}$? Interpret the result.

OR

How much distance has the dolphin covered before hitting the water level again?

4. If α, β, γ are the zeroes of $ax^3 + bx^2 + cx + d$, then $\alpha + \beta + \gamma = -\frac{b}{a}$, $\alpha\beta + \beta\gamma + \gamma\alpha = -\frac{c}{a}$, $\alpha\beta\gamma = -\frac{d}{a}$

(i) If α, β, γ are the zeroes of $x^3 - 5x^2 - 2x + 24$ and $\alpha\beta = 12$ then $\gamma =$

- (a) 2 (b) -2 (c) 3 (d) -3

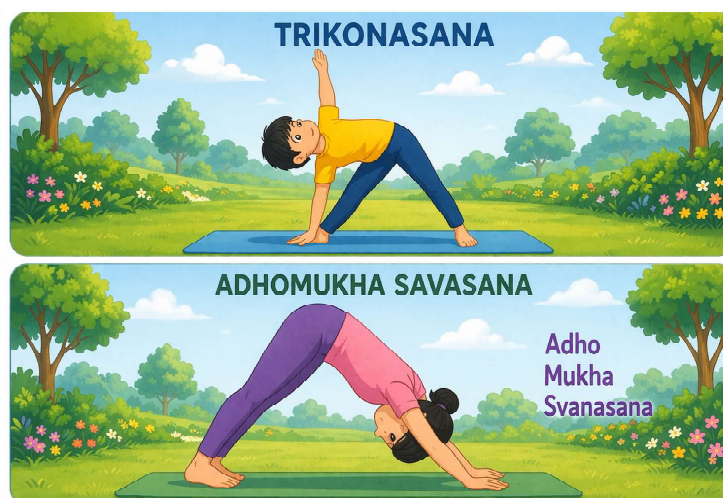
(ii) If $a - b, a, a + b$ are the roots of $x^3 - 3x^2 + x + 1$, then $a + b^2 =$

- (a) 3 (b) 4 (c) 5 (d) 2

(iii) If two zeroes of the polynomial $x^3 - 5x^2 - 16x + 80$ are equal in magnitude but opposite in sign, then zeroes are

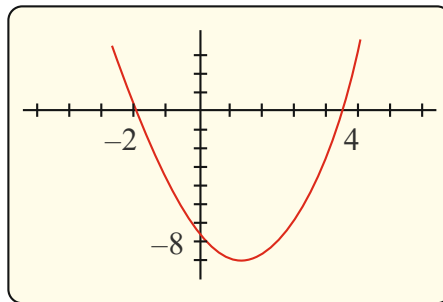
- (a) 4, -4, 5 (b) 3, -3, 5 (c) 2, -2, 5 (d) 1, -1, 5

5. An asana is a body posture, originally and still a general term for a sitting meditation pose and later extended in hatha yoga and modern yoga as exercise, to any type of pose or position, adding reclining, standing, inverted, twisting and balancing poses. In the figure, one can observe that poses can be related to representation of quadratic polynomials. If $4\sqrt{3}x^2 + 5x - 2\sqrt{3}$ represents the quadratic polynomial of Trikonasana.



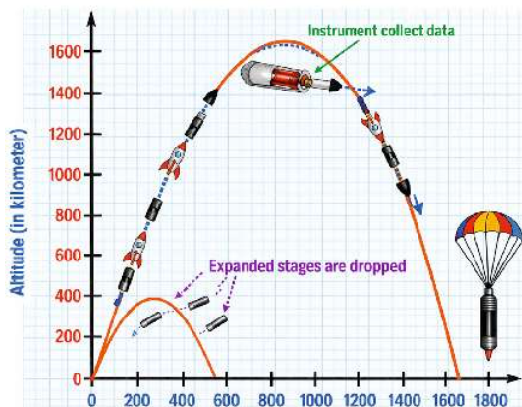
Based on the given information, answer the following questions :

- (a) Name the shape of the poses shown in the figure above
(b) Write the number of zeroes for given graph



(c) Find the zeroes of the quadratic polynomial represent the Trikonasana.

6. SLV-3 was successfully launched on July 18, 1980 from Sriharikota Range (SHAR), when Rohini satellite, RS-1, was placed in orbit, thereby making India the sixth member of an exclusive club of space-faring nations. SLV-3 employed an open loop guidance (with stored pitch programme) to steer the vehicle in flight along a pre-determined trajectory. The successful culmination of the SLV-3 project showed the way to advanced launch vehicle projects such as the Augmented Satellite Launch Vehicle (ASLV), Polar Satellite Launch Vehicle (PSLV) and the Geosynchronous Satellite Launch Vehicle (GSLV).



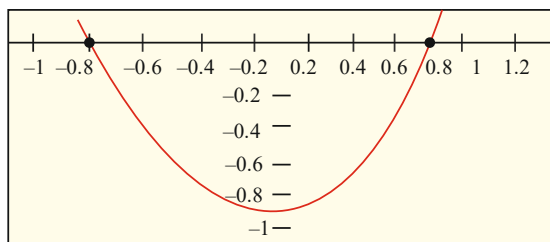
A toy rocket is fired into the air from the top of a tall building. Let its height above the ground after time t seconds be given by $p(t) = 24t^2 - 41t + 12$.

Based on the given information, answer the following questions :

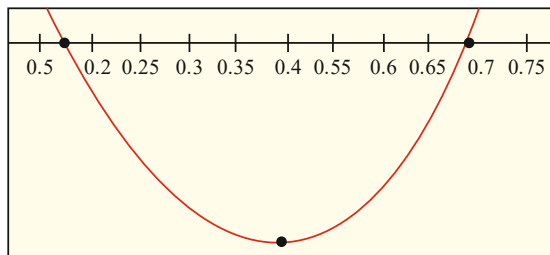
- What will be the shape of the graph representing the height of the rocket above the ground?
 - Find the number of zeroes of the given graph.
 - Find the zeroes of the graph represented by $p(t) = 24t^2 - 41t + 12$.
7. A bakery is trying to find the perfect recipe for their new artisanal bread. The key factor is the rise time of the dough, which can be modeled by a quadratic function $p(x) = 2x^2 - 3x - 1$.

Four bakers Ansh, Ravi, Nitya and Jyoti each propose a graph as follows:

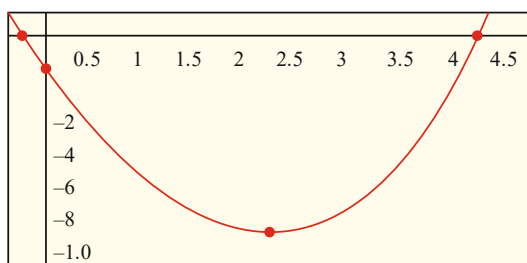
- | | | | |
|----------|----------|-----------|-----------|
| (a) Ansh | (b) Ravi | (c) Nitya | (d) Jyoti |
|----------|----------|-----------|-----------|
- (a) Ansh



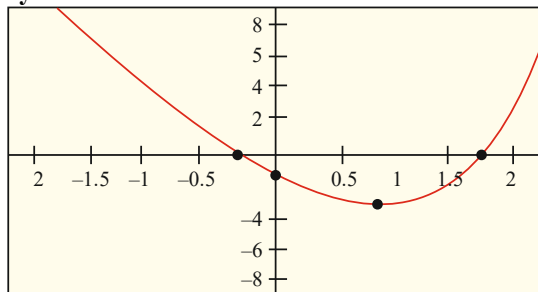
(b) Ravi



(c) Nitya



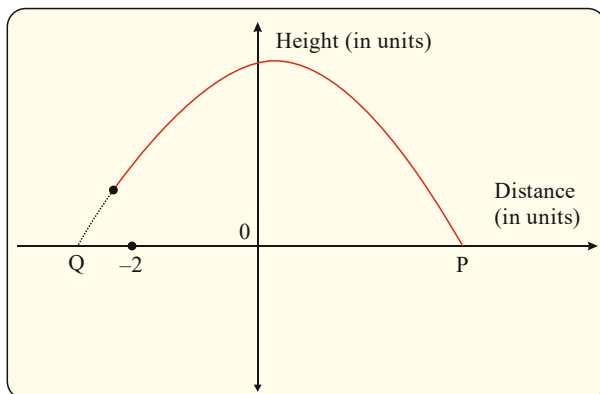
(d) Jyoti



Based on the given information, answer the following questions :

- How many zeros are there for Jyoti's graph?
- Find the zeros of quadratic polynomials. (use $\sqrt{17} = 4.123$)
- How many bakers have drawn the graph correctly?

8. Riddhi throws a stone in the air such that it follows a parabolic path before it lands at P on the ground as depicted by the graph below.



- (a) The above graph represents a polynomial where the sum of its zeroes is 1 and the sum of the squares of its zeroes is 25. Find the coordinates of P and Q.
- (b) If one unit on the graph represents 25 metres, how far from Riddhi does the stone land? Show your work.

Answer Key**Exercise-1****Part - 1 (Basic Multiple Choice Questions)**

- (1) C (2) B (3) A (4) D (5) B (6) A
 (7) C (8) C (9) A (10) B (11) C (12) D
 (13) B (14) D

Part - 2 (Basic Subjective Type Questions)

- (1) (a) 2 (b) 4 (c) 6 (d) 1 (2) Zeroes = $\frac{3}{2}$ and $-\frac{1}{3}$
 (3) a = 1 (4) (i) $= \frac{3}{5}$ (ii) = -5 (5) $36x^2 - 13x + 1$
 (6) (a) $= \frac{-1}{2}$ (b) $= \frac{13}{9}$

**Exercise-2****Part - 1 (Advanced Basic Multiple Choice Questions)**

- (1) C (2) A (3) D (4) A (5) C (6) C
 (7) C (8) B (9) A (10) C (11) A (12) D
 (13) D (14) B (15) A (16) D

Part - 2 (Advanced Subjective Type Questions)

- (1) $2 - \frac{4b^2}{ac} + \frac{b^4}{a^2c^2}$
 (2) $p(x) = x^2 - 10x + 22$
 (3) (i) Number of not polynomials = 3 (ii) Number of quadratic polynomials = 1
 (4) Zeroes = $\frac{-b}{a}, \frac{c}{b}$
 (5) $(3x^2 - 16x + 16)\frac{1}{3}$ (6) $p(x) = k\left(x^2 - \frac{25}{2}x + 41\right)$, where $k \neq 0$

(7) $cx^2 + bx + a$

Part - 3 (Advanced Miscellaneous Questions)

- (1) A (2) B (3) A (4) C (5) A (6) A
 (7) D (8) A

**Exercise-3****(Case Based Questions)**

- (1) (i) Zeroes = 4, -2 (ii) a (iii) The required equation = $x^2 - 36$ **OR**
 (iii) No real value of x is possible, the no of zeroes is 0
- (2) (i) $p(x) = 2x^3 + 4x^2 - 2x + 8$ (ii) Revenue = Rs 50,40,007 and Profit = Rs. 20,39,808
 (iii) Profit generated = Rs. 816.4
- (3) (i) $t = 0, \frac{5}{4}$ (ii) a (iii) $h = -6$ **OR** $d = 25$ cm
- (4) (i) b (ii) a (iii) a
- (5) (a) Parabola (b) Number of zeroes = 2 (c) Zeroes = $\frac{\sqrt{3}}{4}$ and $\frac{-2}{\sqrt{3}}$
- (6) (a) Parabola (b) No. of zeroes = 2 (c) Zeroes = $\frac{3}{8}$ and $\frac{4}{3}$
- (7) d (8) (a) P(4,0) and Q(-3, 0) (b) 150m